

Astra: Histology-Derived Single-Cell Spatial Transcriptomics for Clinical Prediction

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¹Origin Bio

Spatial transcriptomics ties gene expression to tissue architecture, but the assays remain expensive and sparse relative to hematoxylin-and-eosin (H&E) histology, which is abundant and already embedded in pathology workflows. We introduce **Astra**, a model that predicts spatial transcriptomics from H&E at single-cell resolution across diverse tissues. At inference Astra requires only H&E patch embeddings: it predicts the cellular layout within a patch and uses that inferred layout to assign per-cell gene expression, with no external cell segmentation or transcriptomic coordinates. Trained only on the HEST-1k Xenium corpus across breast, lung, bowel, and skin, Astra recovers structured, spatially coherent expression on held-out patients and on an out-of-distribution 10x breast-cancer cohort. We then show that Astra’s H&E-derived expression carries clinical signal without task-specific fine-tuning of the image-to-expression model: it predicts trastuzumab response in HER2-positive breast cancer from intratumoral expression dispersion (AUROC 0.713), stratifies overall survival in ER-positive TCGA-BRCA (log-rank $p = 0.007$ between predicted risk quartiles), and pre-screens for microsatellite instability in colorectal cancer (AUROC 0.705). Together, these results show that histology-derived single-cell expression can support a variety of clinical applications, such as treatment-response prediction, prognosis, and molecular biomarker screening.

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1 Introduction

Spatial transcriptomics reads out gene expression while preserving where each cell sits, so it captures niches, tumor-immune boundaries, and local cell states that bulk or dissociated assays average away [1]. That spatial context is exactly what makes it valuable for understanding disease [2], yet the assays are costly, low-throughput, and sparse. Hematoxylin-and-eosin (H&E) histology is the opposite: it is cheap, ubiquitous, and already sits in every pathology archive. Learning to predict expression directly from H&E would turn that existing morphology into a molecular and spatial readout at cohort scale. That spatial readout matters clinically because many therapies act through cellular geometry rather than average composition: antibody-drug conjugates such as trastuzumab deruxtecan kill antigen-negative neighbors within a short diffusion radius (the bystander effect [3, 4]), so response in antigen-heterogeneous and HER2-low tumors depends on how positive and negative cells are interspersed [5]; likewise, checkpoint-blockade response tracks tumor-T-cell arrangement and organized tertiary lymphoid structures [6–8]. Each is a property of tissue geometry that survives only when a cell’s position is recorded alongside its expression.

Existing models establish that morphology carries transcriptomic signal, but their resolution or inference requirements limit how directly they can be used. Spot-level models such as ST-Net, HisToGene, and Hist2ST [9–11] learn from Visium-style targets, so each prediction averages over many cells and blurs the cell-to-cell variation that defines a niche. Single-cell methods such as GHIST and DeepCell improve resolution but still take the cellular layout as an input rather than inferring it: GHIST matches separate H&E and Xenium segmentations, while DeepCell crops H&E around supplied Xenium centroids [12, 13]. Either route ties training and inference to an external

segmentation, and any spatial mismatch between the two assays injects label noise or discards usable training signal (Section 3).

Astra removes that dependency. At training, cells enter the model as queries defined by their centroids, while, in parallel, a density head learns where those centroids fall inside each patch. At inference, the density head proposes the layout from H&E alone and expression is queried at the inferred positions, without external segmentation or cross-assay matching. We first test whether this H&E-only pipeline recovers spatially coherent expression across tissues. We then hold Astra frozen and test three clinical applications in cohorts without measured spatial transcriptomics: trastuzumab response prediction in HER2-positive breast cancer, survival stratification in ER-positive breast cancer, and MSI-H pre-screening in colorectal cancer. These tasks test whether the predicted transcriptome provides useful clinical signal rather than only reproducing benchmark expression.

2 Astra recovers spatial expression from histology

We evaluate Astra with patient-level cross-validation on HEST-1k Xenium data restricted to breast, lung, bowel, and skin, withholding entire patients each fold. The primary metric is the per-gene Spearman correlation between predicted and measured expression (Methods 8.6): cells with higher measured expression should rank higher in the prediction, which, because every cell has a coordinate, is equivalent to recovering where a gene is high or low on the slide. Genes vary widely in how much spatial structure they carry. Diffuse genes are nearly random across the tissue and hard to pin to morphology; genes that organize into compartments or neighborhoods are the ones Astra recovers (Figure 1a). Across the top-50 Moran’s I genes the median per-gene Spearman is 0.46 in breast, 0.51 in lung, 0.41 in bowel, and 0.53 in skin (0.47 pooled), a consistent level of recovery across four unrelated tissue types. Ranking each slide’s genes by true Moran’s I and sweeping how many are kept, this recovery is highest for the most organized genes and falls as more diffuse genes are added (Figure 1b), so the top few dozen genes per slide carry the strongest predictions.

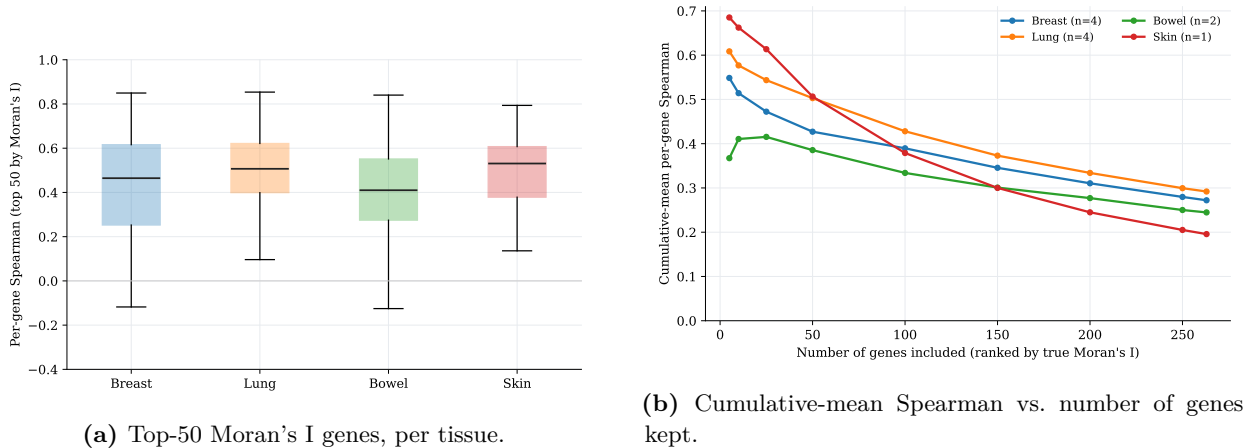


Figure 1: Held-out HEST-1k recovery with inferred layouts: top-50 gene Spearman by tissue and cumulative recovery across Moran-ranked genes.

Because Astra is trained only on HEST-1k, a useful stress test is an out-of-distribution cohort with different acquisition and batch characteristics. On 10x Genomics’ public BreastCancer1 and BreastCancer2 Xenium slides, the per-gene Pearson correlation for the top 25 genes by Moran’s I has a median above 0.80 in both cohorts, with the top 50 genes near 0.74 (Figure 2). These

medians meet or exceed the values GHIST reports on the same two slides (top-20 spatially variable genes ≈ 0.70 , top-50 ≈ 0.60) [12], and Astra reaches them fully out of distribution, having never seen these slides, whereas GHIST is trained on each slide through five-fold spatial cross-validation over horizontal bands of the same tissue. The comparison is not head-to-head, since the gene rankings differ (Moran’s I vs. SPARK-X spatially variable genes) and our correlation is computed over all cells on a slide rather than a single held-out band, but it indicates Astra’s out-of-distribution recovery is competitive with a model fit directly on the target slides.

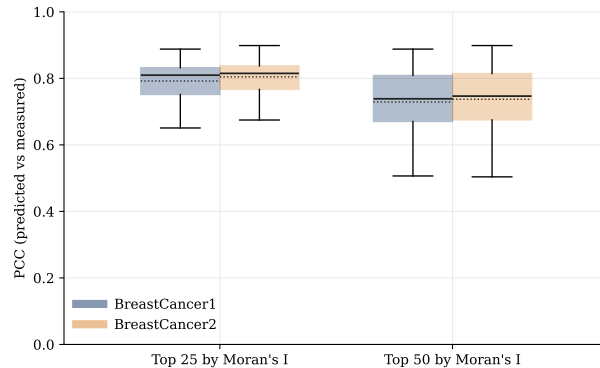


Figure 2: Out-of-distribution 10x breast PCC for the top 25 and 50 genes ranked by Moran’s I.

Aggregate correlations do not show what the predictions look like in tissue. We therefore examine one biologically informative gene from each tissue, always showing measured expression beside Astra’s prediction.

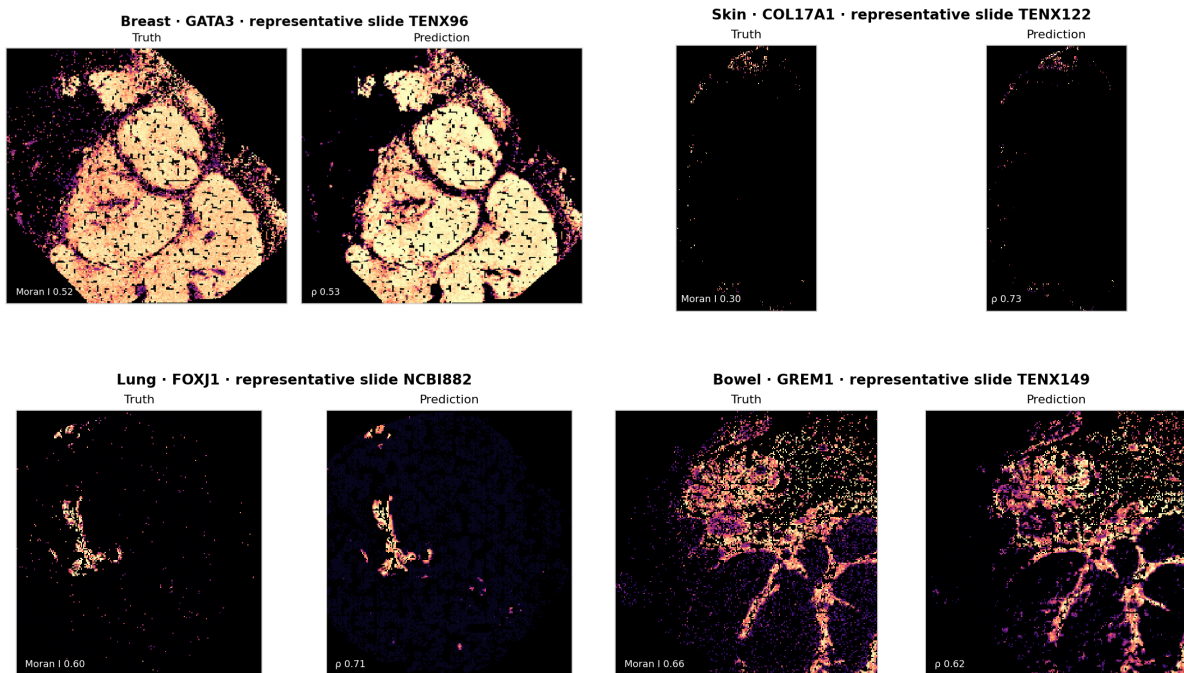


Figure 3: Measured (left within each panel) and Astra-predicted (right) expression for GATA3 in breast, COL17A1 in skin, FOXJ1 in lung, and GREM1 in bowel.

These maps test whether Astra places high-expression domains and compartment boundaries correctly rather than only matching a slide-level distribution, spanning four distinct morphology-expression relationships. GATA3 marks luminal mammary epithelium and estrogen-receptor signaling, a defining regulator of luminal breast-cancer identity [14]; COL17A1 anchors basal ker-

atinocytes through hemidesmosomes and marks a distinct basal skin compartment [15]; FOXJ1 drives motile-cilia differentiation, localizing ciliated airway epithelium against surrounding lung [16]; and GREM1, a BMP antagonist, marks a cancer-promoting stromal program in colorectal tissue [17]. Astra recovers these domains with Spearman correlations of 0.53, 0.73, 0.71, and 0.62, respectively. These are qualitative spatial sanity checks chosen to cover distinct compartments (a nuclear luminal regulator, a basal adhesion protein, a ciliogenesis factor, and a secreted stromal antagonist), not the highest-correlation genes or evidence that they are privileged biomarkers; the held-out aggregate analyses (Figure 1) provide the quantitative assessment.

3 Architecture and data

Astra uses a frozen H&E image encoder to convert each tissue patch into visual tokens, followed by a shared transformer trunk that integrates morphology across the patch (Methods 8.1). From this representation the model branches into three heads (Figure 4): a per-cell expression head, a detector head that predicts whether each target gene is non-zero in a given cell, and a density head that predicts where cells lie within the patch.

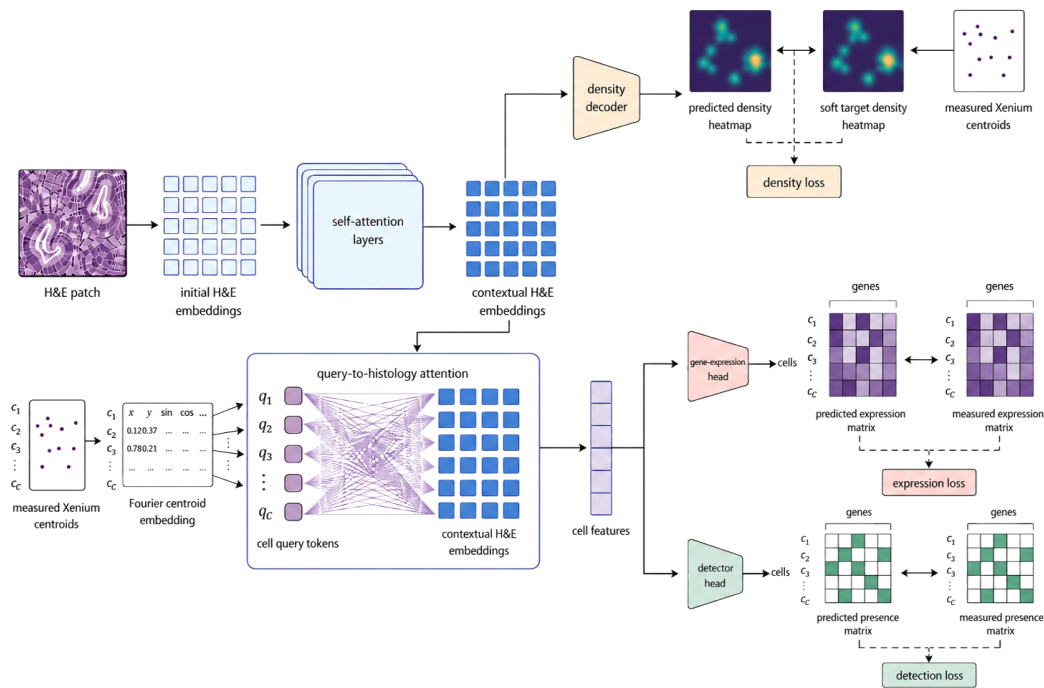


Figure 4: Astra’s shared visual trunk and position-conditioned density, presence, and per-cell expression heads.

The HEST-1k Xenium slides do not share a single gene panel: the breast, lung, bowel, and skin assays each target a different set of transcripts. We reconcile these per-tissue Xenium panels into one harmonized vocabulary and train Astra to predict this fixed 1,656-gene output for every cell, regardless of which tissue the patch comes from. This single output space is what the downstream clinical analyses use: in Sections 4–6 each patient is run through the full 1,656-gene head, since those tasks operate per patient and never require a common gene set across Xenium slides. The per-slide recovery in Section 2 reports on far fewer genes: each slide is scored on its own top genes ranked by that slide’s Moran’s I, so the actual genes differ from slide to slide. To average a top- K statistic over all held-out slides on equal footing, every slide must be able to supply K genes, which

caps K at the gene count of the smallest-panel slide, a few hundred genes. The number thus differs by application because of how the metric is aggregated, not because the model emits a different panel.

The central design decision is that cells enter as queries defined by their own centroids. Each cell’s relative position inside the patch is encoded with a Fourier embedding and used as a query that cross-attends to the patch’s image tokens, producing a feature for that specific cell. This avoids the object mismatch of methods that detect nuclei on H&E and then match them to a transcriptomics segmentation: Astra has no matching step, because the centroids themselves are the queries.

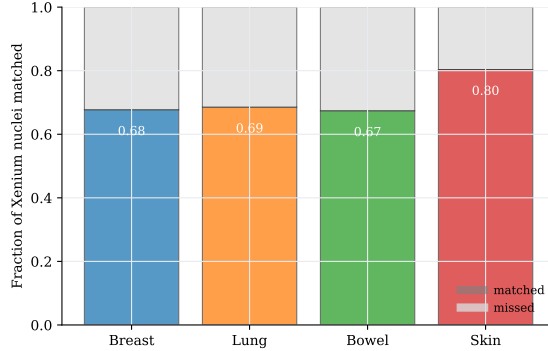


Figure 5: Xenium nuclei matched to H&E detections by tissue; grey, unmatched.

This is a design decision about how training supervision is used, not a benchmark against matching-based methods. H&E detections and Xenium nuclei are non-identical object sets: nearest-neighbor matching recovers 68% of Xenium cells in breast, 69% in lung, 67% in bowel, and 80% in skin, leaving 20–33% unmatched (Figure 5). Matched-pair supervision must therefore discard or misassign part of the measured transcriptome, whereas Astra anchors training queries to measured Xenium centroids and uses every measured cell, estimating their positions only at inference.

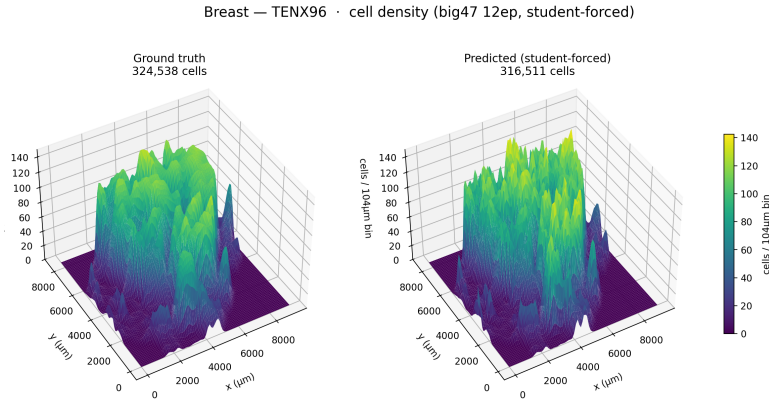


Figure 6: Measured and Astra-decoded density on held-out breast (104 μm bins; 324,538 vs. 316,511 cells).

The density head is what makes inference H&E-only. During training, cell queries are placed at the true centroids while the density head learns the cellular layout from a unit-mass Gaussian target (Methods 8.2); at inference Astra proposes that layout and queries expression at the inferred positions. At inference the head emits a continuous density heatmap that must be turned into discrete cell positions. We decode it adaptively (Methods 8.5): because the density target is built from unit-mass Gaussians, integrating the map gives a cell count C , and comparing C against the number of resolvable local maxima says which decoder to trust. Where peaks account for at least 80% of C we take those non-maximum-suppression peaks directly; where the density saturates

and peaks merge, so that fewer than 80% of the predicted cells appear as distinct maxima, we instead place all C cells as density-weighted centroids by a Lloyd/centroidal-Voronoi relaxation, which recovers the correct cell count in crowded regions where peak picking would undercount. The decoded layout closely tracks the measured one both in total count and in spatial distribution (Figure 6).

Because the decoded and measured cells are two independent point sets, we pair them for scoring by greedy nearest-neighbor matching (Methods 8.6): processed in increasing distance, each measured cell claims its closest unclaimed predicted cell within a distance gate set by the slide’s typical cell spacing, giving one-to-one pairs (unmatched cells count as misses or false positives), and per-gene Spearman correlations are computed over these matched pairs. To isolate the cost of inferring rather than measuring the layout, we re-score the held-out slides with either measured centroids (*teacher-forced*) or Astra’s decoded layout (*student*); the gap between the two is small (Figure 7).

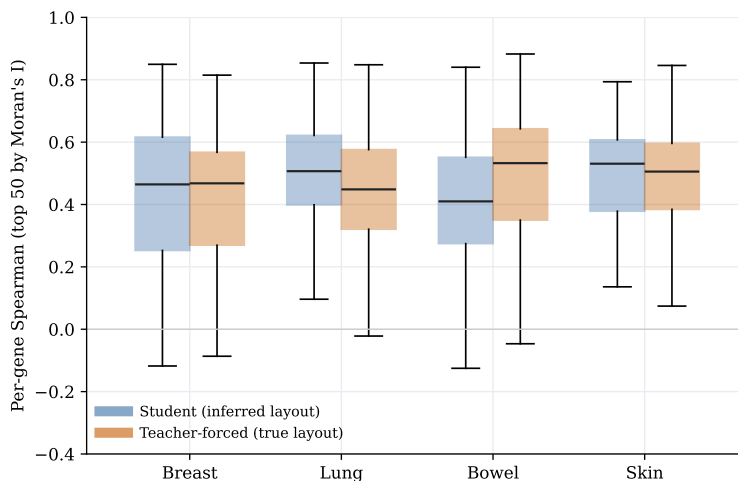


Figure 7: Top-50 gene Spearman with inferred (student) versus measured (teacher-forced) centroids.

The student model recovers almost all teacher-forced performance without measured centroids or an external cell localizer. Median Spearman is 0.46 versus 0.47 in breast and 0.53 versus 0.51 in skin, while lung is similarly close. Bowel shows the only visible gap. The overlap across tissues indicates that layout inference preserves the expression signal in the full H&E-only pipeline.

The detector head addresses a different failure mode. A continuous head trained with mean-squared error on $\log(1+x)$ never outputs hard zeros, so absent genes emerge as small non-zero values and can fabricate infiltration for markers such as PTPRC [18]. A separate head predicts whether each gene is non-zero, and gating by that probability restores true absences. One fixed threshold, selected on held-out validation slides, is used throughout. The detector reaches AUROC 0.96 in breast and lung, 0.94 in skin, and 0.87 in bowel (pooled 0.94; Figure 8).

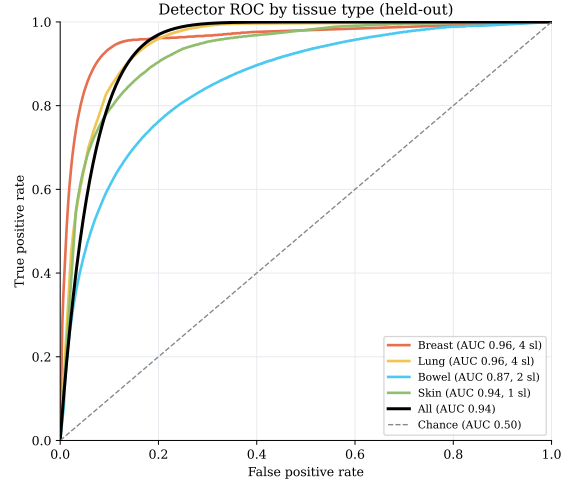


Figure 8: Held-out gene-presence detector ROC by tissue.

To sanity-check the training mixture and decide which tissues to keep, we trace each held-out query patch back to the training patches that most helped or hurt its prediction with influence functions. For a query patch q and a training candidate c the FastInf score is $I(c, q) = -(\nabla_{\theta} L_c)^{\top} \hat{G}_{\lambda}^{-1} (\nabla_{\theta} \ell_q)$, where ℓ_q is the all-genes MSE loss on the held-out patch and L_c the same loss on the candidate. The curvature term $\hat{G}_{\lambda}$ is a block-diagonal generalized Fisher matrix accumulated over a Fisher subsample of $N_F = 1,024$ training patches (64 batches of 16) as $\hat{G} \approx \frac{1}{N_F} \sum_i g_i g_i^{\top}$ per trainable matrix block (a diagonal Fisher for vector blocks), then *damped* by adding a multiple of the identity before inversion,

$$\hat{G}_{\lambda} = \hat{G} + \lambda I,$$

and inverted with a Schulz hyperpower iteration (HyperINF). The damping term λ is what makes the inverse well-conditioned: the empirical Fisher is low-rank and near-singular at this subsample size, so λ floors its spectrum and bounds $\|\hat{G}_{\lambda}^{-1}\|$, trading curvature fidelity for numerical stability. We use $\lambda = 0.01$, the smallest value in a $\lambda \in \{0.01, 0.05, 0.1\}$ sweep that still inverts stably, since larger damping drives $\hat{G}_{\lambda}^{-1} \rightarrow \lambda^{-1} I$ and degrades the score toward a plain gradient dot product. The score is signed: a negative value marks a beneficial (proponent) patch that lowers the query’s loss, a positive value a harmful (opponent) one. Scoring each query against the whole training set is intractable, so we narrow candidates in two stages. First a FastIF-style retrieval step picks *which* patches are eligible: for each query we take its $k = 10,000$ nearest training patches by L2 distance between frozen Gigapath class-token features, a purely visual-similarity filter. We then compute the influence score above for that shortlist and rank patches by the score itself, so “most influential” is decided by influence, not by feature distance. We report raw (non-cosine) scores at $\lambda = 0.01$ and $k = 10,000$ over 1,024 held-out queries.

The recovered relationships pass an obvious sanity check and surface a useful data decision. Beneficial influence is overwhelmingly same-tissue and morphologically matched: a query’s strongest proponents are training patches of the same tissue that share its morphology (same-tissue fraction 0.96 lung, 0.92 breast, 0.86 pancreas, 0.84 skin; Figure 9). The systematic exception is bowel, whose beneficial influencers are only 0.58 bowel and draw heavily on pale fibrous/stromal patches from pancreas (0.26) and breast (0.08), so influence tracks shared morphology, not the tissue label. That is what tells us to include the pancreas slides even though Astra is not intended for pancreas inference: pancreas is the single largest cross-tissue source of beneficial signal for bowel, a tissue we do report on. The qualitative panel shows the same pattern patch by patch, with each query

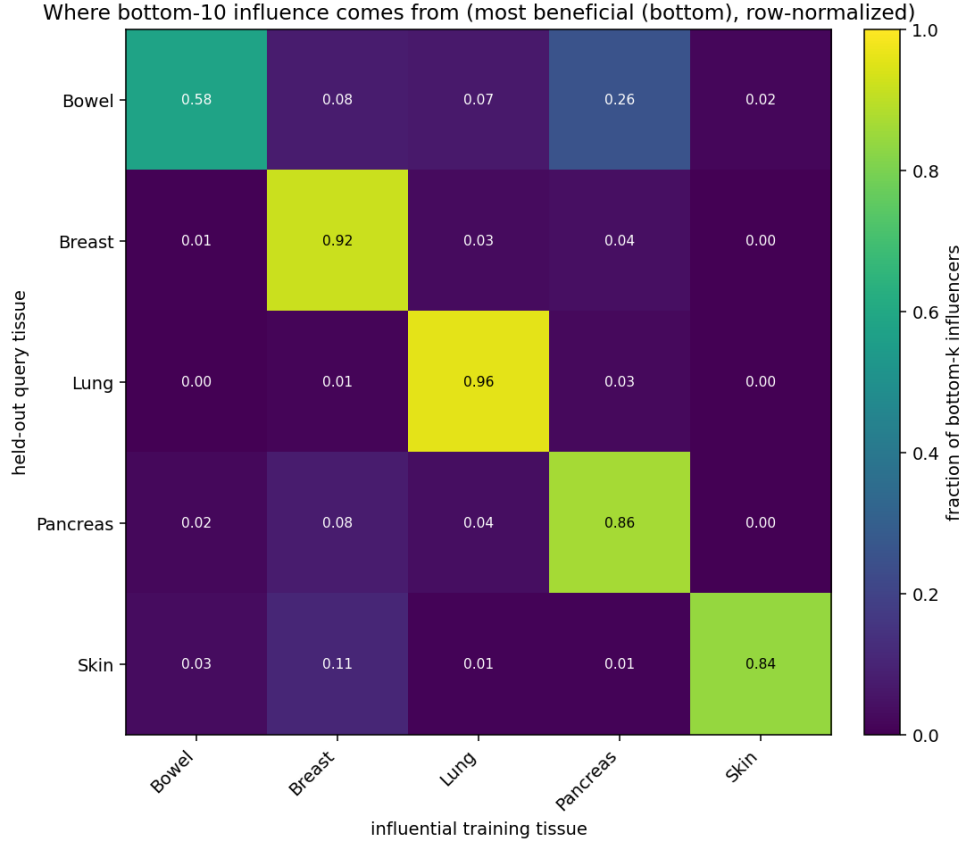


Figure 9: Where beneficial (bottom-10) influence comes from, row-normalized by held-out query tissue. Proponents are dominated by the same tissue, with bowel the exception (strong pancreas contribution).

retrieving same-tissue twins as its top proponents while bowel pulls in breast and pancreas stroma (Figure 10).

The magnitude of these scores is worth reading carefully, because ranks here are defined by influence rather than by feature distance (Figure 11). Lists are sorted by score descending, so rank 1 is a query’s *least* beneficial scored candidate. The median of those rank-1 scores is -3.8 (left), already on the beneficial side of zero, which means that for a typical held-out query essentially the entire scored shortlist is signed beneficial and outright harmful candidates are the exception. The right panel shows the median and 10–90 percentile band of the score at each rank position once a query’s 10,000 candidates are sorted. It is therefore a sorted quantile profile of one distribution: its monotone descent is guaranteed by the sorting and carries no information about feature-space proximity or about how deep a kNN pool needs to reach. What it does report is the shape of that distribution. The bulk of candidates sit in a shallow band around the median, while the final decade of ranks steepens sharply, the median reaching about -22 and the 10th percentile falling below -60 . Strongly beneficial influence is thus concentrated in a small minority of patches rather than spread evenly over the shortlist, which is why the tissue attributions above are read from the bottom-10 proponents per query.

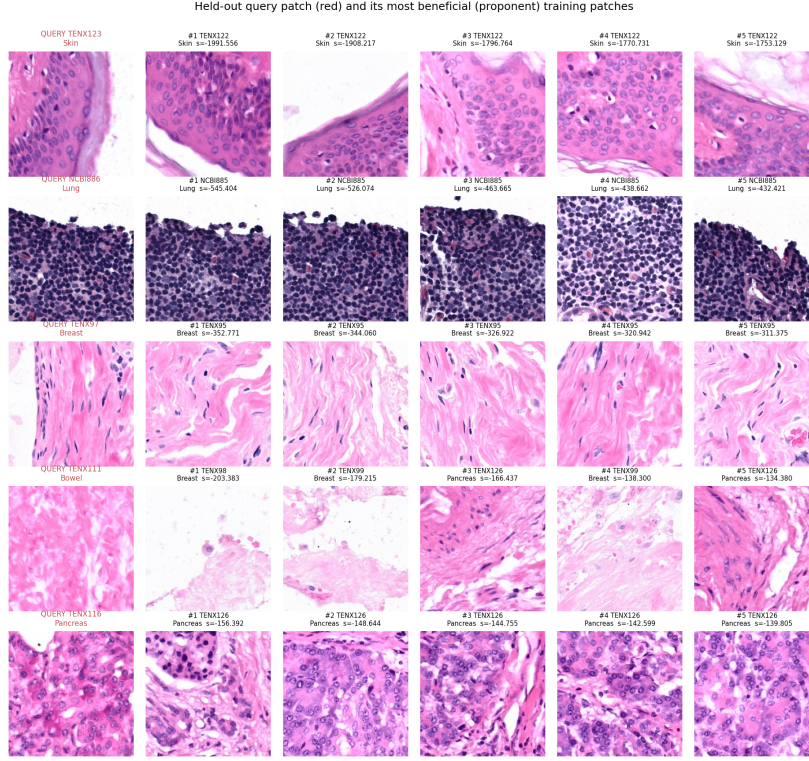


Figure 10: Held-out query patch (red, leftmost) and its top beneficial proponent training patches per row; same-tissue morphological twins dominate, while bowel retrieves breast and pancreas stroma.

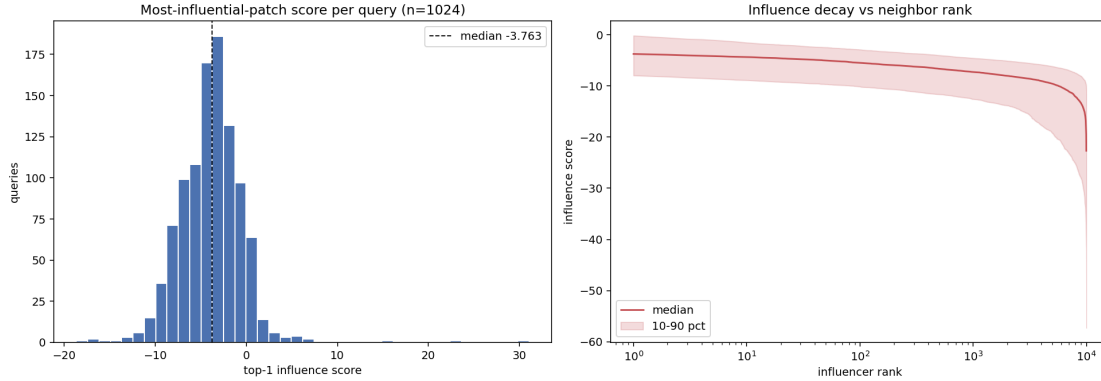


Figure 11: Influence score geometry over 1,024 held-out queries ($\lambda = 0.01$, $k = 10,000$, raw scores). Left: distribution of each query’s rank-1 (least beneficial) influence score, median -3.8 , already on the beneficial side of zero. Right: median and 10–90 percentile band of the score at each rank position after sorting each query’s candidates by influence. Ranks are defined by the score itself, so this is a sorted quantile profile of the score distribution and its monotone descent is a property of the sorting, not evidence about feature-space proximity; the sharp steepening in the last decade shows that strongly beneficial influence is confined to a small minority of candidates.

4 Astra predicts trastuzumab response

Having shown that Astra recovers spatial expression from histology, we now hold it frozen and ask whether that H&E-derived expression carries clinical signal in cohorts where spatial transcriptomics was never measured, beginning with treatment response. We use trastuzumab (anti-HER2) response in HER2-positive breast cancer on the Yale cohort (85 pre-treatment patients, 36 responders and 49 non-responders), labeled by pathologic response versus residual disease [19]. Astra stays frozen. Within each patient’s tumor ROIs (Methods 8.4) it detects cells and predicts their expression tile by tile, giving a stack of up to roughly 512 tiles across the 1,656-gene output panel per patient.

Rather than averaging expression away, we characterize how each gene varies across the tumor. For every gene we compute two dispersion statistics across the patient’s tiles (Methods 8.7): the tile-to-tile standard deviation and the gap between the 90th percentile and the median (a hotspot term). This yields 3,312 features per patient describing the spatial heterogeneity of the predicted transcriptome, without directly including mean expression. In each outer 5-fold split, the 17 test patients remain untouched. The feature count is chosen by an inner stratified 5-fold cross-validation over the 68-patient development set: for each candidate count from 1 through 20 we standardize and rank features by response correlation on each inner training split, fit an L2-regularized logistic regression on the top-ranked features, and score the held-out inner folds; the count with the highest pooled inner-fold AUROC is selected. We then refit the selected model on all 68 development patients before predicting the 17 test patients.

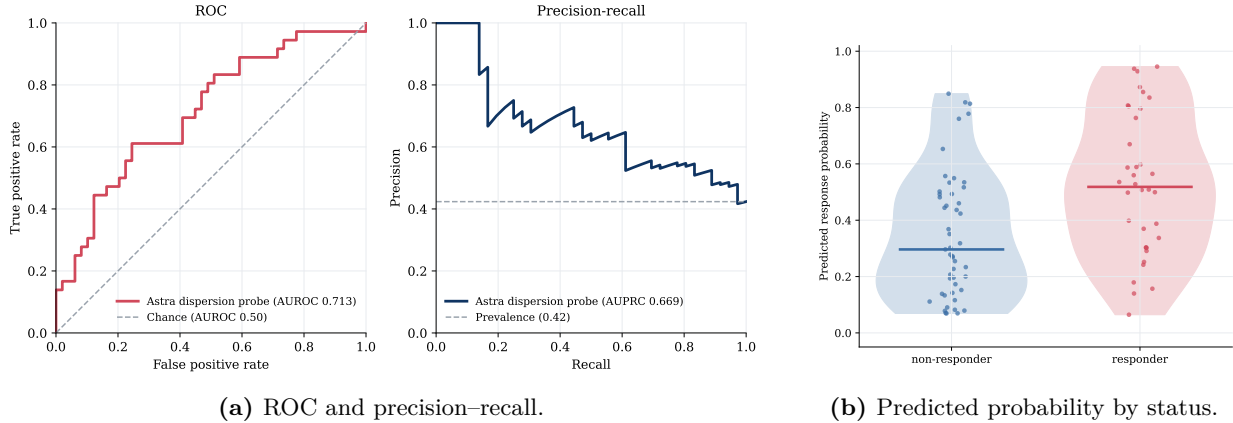


Figure 12: Trastuzumab response from Astra-derived dispersion features ($n = 85$): ROC/PR (AUROC 0.713; AUPRC 0.669) and out-of-fold probabilities.

Evaluated with repeated stratified 5-fold cross-validation (50 repeats) on averaged out-of-fold probabilities, this nested feature-count selection reaches AUROC 0.713 (95% CI 0.60–0.82 by $2000 \times$ bootstrap) and AUPRC 0.669 against a 0.42 prevalence baseline (Figure 12a). The inner cross-validation selected a median of 8 features. Responders receive a higher median predicted probability (~ 0.52) than non-responders (~ 0.30). The difference from prior work is one of supervision: on this same cohort, Farahmand et al. trained an end-to-end Inception-v3 CNN directly on the response label and reported roughly 0.80 slide-level AUROC [19], whereas Astra’s image-to-expression model was optimized only to predict expression, without task-specific fine-tuning on the response label; only the lightweight linear probe is trained on it.

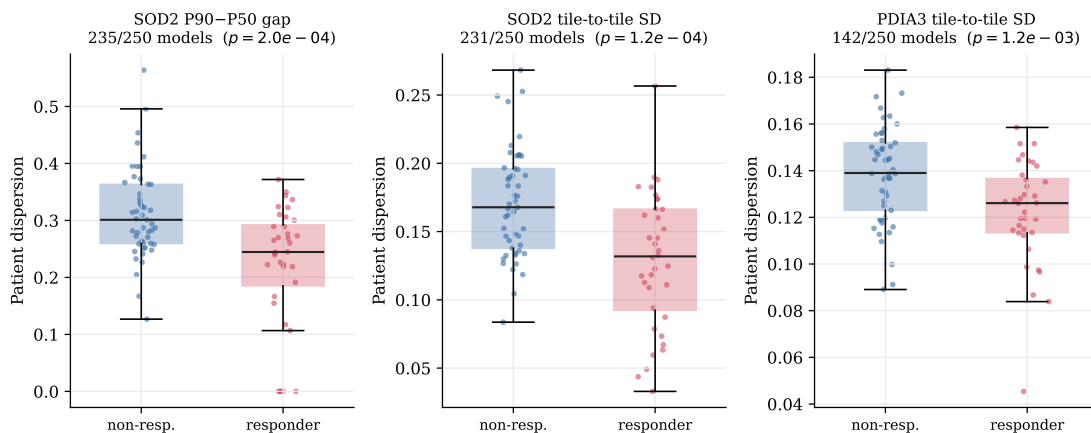


Figure 13: Cohort distributions of the three most selected response features; counts denote selection across 250 outer models.

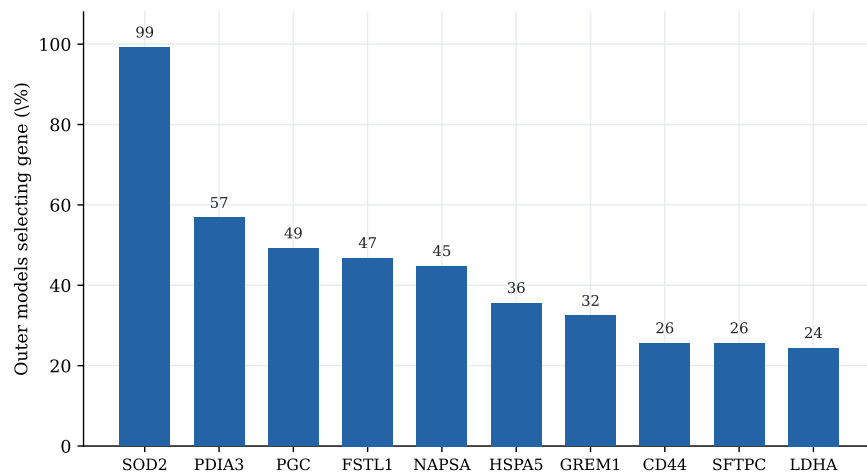


Figure 14: Top-10 genes by union selection frequency: fraction of the 250 outer models selecting either the tile-to-tile standard deviation or the P90–P50 hotspot gap of each gene (the two variants are counted once per model).

Because the classifier is linear and its selection occurs entirely inside the training data, we can inspect which features remain stable without using test labels. The 90th-percentile-minus-median hotspot gap and tile-to-tile standard deviation of SOD2 are selected in 235 (94%) and 231 (92%) of the 250 outer models, respectively; counting either dispersion variant, SOD2 enters 248 of 250 models (99%). The third most frequently selected feature is the tile-to-tile standard deviation of PDIA3 (142 models, 57%; Figures 13–14). SOD2 is the mitochondrial superoxide dismutase that buffers oxidative stress. Elevated SOD2 has been associated with tumor stage, nodal metastasis, and worse survival in breast cancer, and redox regulation has been implicated more broadly in HER2-positive disease and treatment response [20, 21]. PDIA3 (ERp57), the third-ranked feature’s gene, is an endoplasmic-reticulum disulfide oxidoreductase involved in protein folding and redox quality control. Prior breast-cancer studies report increased PDIA3 in invasive ductal carcinoma, including association with high-grade and node-positive disease, while perturbing PDIA3 alters extracellular-matrix, adhesion, and migration programs linked to breast-cancer outcomes [22, 23]. This makes heterogeneous PDIA3 expression biologically plausible as a regional stress or aggressive-tissue marker, but does not establish it as a trastuzumab-resistance mechanism.

Beyond these two, the remaining recurrently selected genes (Figure 14) fall into three tiers of literature support (Table 1): three with direct trastuzumab-resistance evidence, three with general

breast-cancer associations but no trastuzumab-specific role, and two lung-lineage markers with no established breast role that most plausibly reflect predicted-expression variation the probe latches onto rather than a mechanism. This audit supports biological plausibility for the leading genes but is not independent validation, so the quantitative cohort test that follows is limited to the top two features, SOD2 and PDIA3.

Table 1: Recurrently selected trastuzumab-response genes beyond SOD2 and PDIA3, grouped by strength of literature support.

Gene	Role	Literature Context
<i>Direct trastuzumab-resistance evidence</i>		
HSPA5 (GRP78)	UPR chaperone	Zhang & Zhang (2013) [24]
CD44	stem-cell marker	Boulding et al. (2015) [25]
LDHA	glycolytic enzyme	Zhao et al. (2011) [26]
<i>General breast-cancer association</i>		
PGC	pepsinogen C	Balbin et al. (1995) [27]
FSTL1	secreted glycoprotein	Chan et al. (2018) [28]
GREM1	stromal BMP antagonist	Kobayashi et al. (2021) [17]
<i>No established breast role (likely expression proxy)</i>		
NAPSA	lung-adenocarcinoma marker	Turner et al. (2012) [29]
SFTPC	surfactant protein C	Turner et al. (2012) [29]

For both features, the direction is consistent across the cohort: both SOD2 dispersion terms are higher in non-responders (standard deviation 0.171 vs. 0.129, $p = 1.2 \times 10^{-4}$; hotspot gap 0.312 vs. 0.222, $p = 2.0 \times 10^{-4}$), as is PDIA3 standard deviation (0.138 vs. 0.122, $p = 0.0012$; Figure 13). Non-responding tumors therefore carry patchier oxidative- and ER-stress predictions. These statistics retain variation that pseudobulk erases, although their correlation with mean abundance prevents a claim of an independent heterogeneity mechanism. Figure 15 localizes that variation: the SOD2 hotspot gap reflects high-expression tiles against a lower background, whereas PDIA3 standard deviation captures broader tile-to-tile variability. These within-group maxima illustrate the phenotype; the cohort evidence is Figure 13.

Response here was predicted without any trastuzumab-specific training of the expression model: the same frozen Astra supplies the features and only a small linear probe is fit on the labels. This is the point of routing clinical prediction through predicted expression rather than training a bespoke supervised model per task. That expression is a shared substrate, so as the base model improves from aggregated spatial transcriptomics across more patients, its predictions become more accurate and every downstream readout built on them (response here, survival and MSI in the sections that

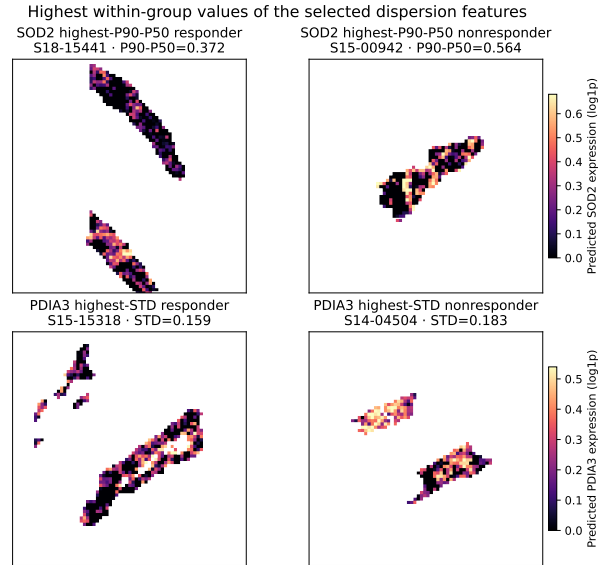


Figure 15: Spatial maxima for SOD2 hotspot gap (top) and PDIA3 standard deviation (bottom); squares are 128 μm H&E tiles.

follow) improves at once, with no new task-specific labels. The clinical results that follow lean on exactly this transfer.

5 Astra predicts survival risk in luminal breast cancer

Trastuzumab response used Astra’s expression to predict a treatment endpoint; we next ask whether the same frozen model carries prognostic signal for a different clinical outcome and cohort. Overall survival in luminal breast cancer is a natural test: molecular expression stratifies risk in ways morphology alone does not make obvious [30], yet measuring it at single-cell spatial resolution is rarely feasible at cohort scale. On 771 ER-positive TCGA-BRCA patients with matched diagnostic H&E and overall survival (97 deaths, 674 censored; median follow-up ~ 865 days), we retain the 1,000 most tissue-dense tiles per patient (Methods 8.4), run frozen Astra to predict a 1,656-gene profile at each decoded cell, and average those predictions into one pseudobulk vector (Methods 8.7). Repeated 5-fold cross-validation (50 repeats) ranks genes by survival association within each training fold and fits a multivariable Cox model; the number of genes is itself chosen per fold by an inner 3-fold cross-validation over the development patients (median 40 genes), and only held-out patients are scored, yielding a pooled concordance index of 0.570.

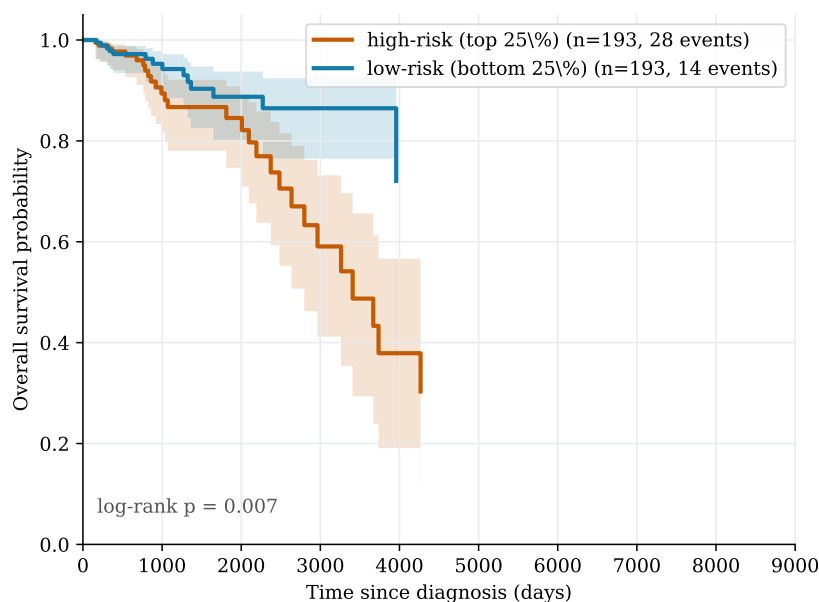


Figure 16: ER-positive TCGA-BRCA survival for the top and bottom cross-validated Astra risk quartiles ($n = 193$ each; two-sided log-rank $p = 0.007$).

The separation is sharpest at the extremes of predicted risk: comparing the top and bottom risk quartiles (193 patients each), the high-risk group has twice as many deaths as the low-risk group (28 vs. 14) and the survival curves diverge early and stay apart (Figure 16; two-sided log-rank $p = 0.007$). Splitting the full cohort at the median risk preserves the separation across all 771 patients (386 high vs. 385 low; 57 vs. 40 deaths; Figure 17; $p = 0.031$).

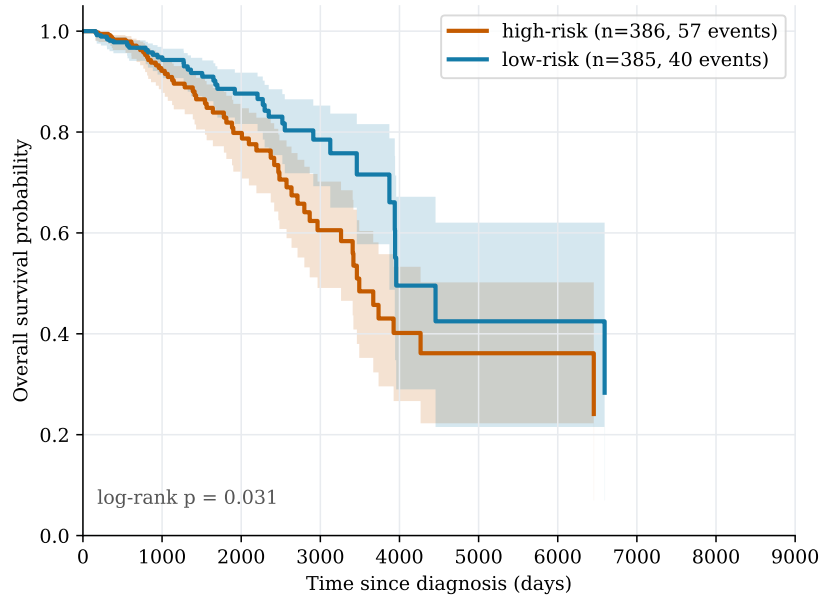


Figure 17: ER-positive TCGA-BRCA survival by cross-validated Astra risk at the median split ($n = 386$ high, 385 low; two-sided log-rank $p = 0.031$).

Because gene selection happens entirely inside the training folds, we can read off which genes recur across the 250 outer models without touching held-out labels (Figure 18). The most frequently selected genes are secretory and stromal (SLPI 86%, CTSE 78%, SERPINB2 73%, PCOLCE2 52%) mixed with a strong adaptive-immune and lymphocyte signature (IGKC 71%, IGHG1 63%, ICOS 60%, PDCD1 59%, TRBC2 57%, CTLA4 51%, MARCO 50%), indicating that Astra recovers a coherent immune- and stroma-linked survival signal from H&E alone.

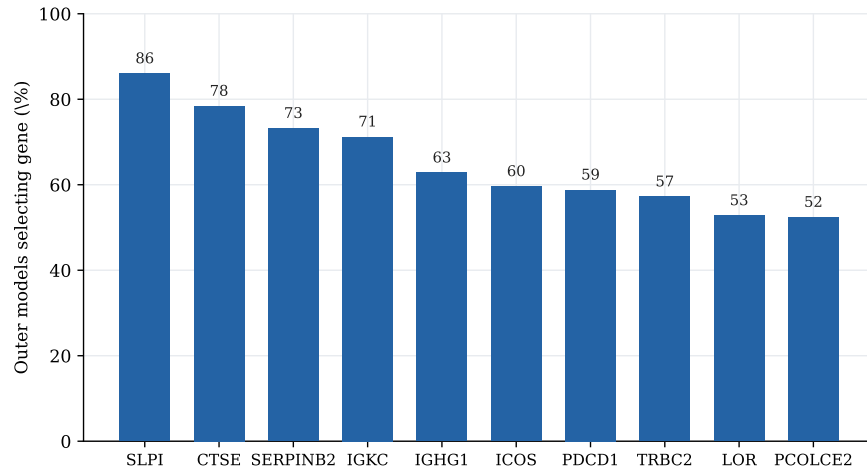


Figure 18: Top-10 genes by cross-validated selection frequency in the luminal survival model (fraction of the 250 outer Cox models retaining each gene).

These recurrent genes have documented breast-cancer associations that match the direction Astra finds. The humoral- and lymphocyte-immunity markers are the best established: immunoglobulin kappa C (IGKC) is a validated single-gene prognostic and predictive marker whose high tumor expression, reflecting infiltrating plasma cells, tracks with better survival in breast cancer [31], and dense tumor-infiltrating lymphocytes marked by IGHG1, ICOS, PDCD1, TRBC2, and CTLA4 are a recognized favorable-prognosis axis in ER-negative and high-grade disease [32]. Among the secretory and stromal genes, SLPI is a driver of triple-negative breast-cancer metastasis whose

high expression predicts shorter overall and metastasis-free survival [33]; cathepsin E (CTSE) is a Mendelian-randomization-linked breast-cancer-risk gene with subtype-dependent prognostic value [34]; and SERPINB2 (PAI-2) marks the plasminogen-activator axis, where high tumor-cell expression is associated with lymph-node metastasis and reduced survival [35]. That an unsupervised probe on H&E-predicted expression rediscovers this immune–stroma prognostic program, rather than arbitrary genes, supports the biological plausibility of the signal while the concordance and log-rank statistics remain the quantitative evidence.

6 Astra can help pre-screen for MSI-H

Microsatellite instability (MSI-H) is among the most actionable biomarkers in oncology: it predicts response to immune-checkpoint inhibitors. Pembrolizumab is FDA-approved for MSI-H/dMMR solid tumors in a tissue-agnostic indication [36], and MSI-H also flags Lynch syndrome for genetic screening. Standard testing relies on PCR or immunohistochemistry for mismatch-repair proteins, so calling MSI from routine H&E would be a cheap pre-screen, and prior deep-learning work has shown histology carries this signal [37]. Our test is deliberately strict: the image-to-expression model receives no task-specific fine-tuning on MSI. It is trained only on HEST-1k spatial transcriptomics, with no TCGA data and no MSI labels anywhere in that model; only the linear probe on top sees the label. We run frozen Astra over TCGA colorectal slides (TCGA-COAD and TCGA-READ; 255 patients, 85 MSI-H), pool predicted per-cell expression into one pseudobulk profile per patient, and fit only a linear classifier on top under nested cross-validation (Methods 8.7; repeated stratified 5-fold outer CV, 50 repeats), with the number of AUC-ranked genes chosen by an inner 5-fold loop inside each training split (median 160 genes), scored against the genomic MANTIS label [38].

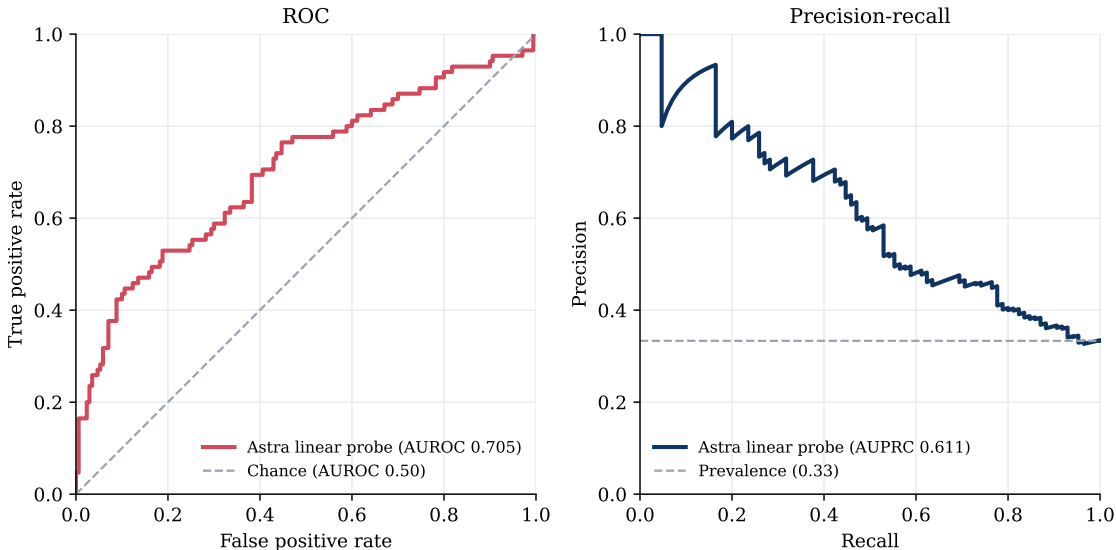


Figure 19: MSI-H detection from Astra-predicted pseudobulk expression in TCGA-COAD/READ (AUROC 0.705; AUPRC 0.611).

The linear probe reaches AUROC 0.705 and AUPRC 0.611 at a cohort prevalence of 33% MSI-H (Figure 19), a 1.8 $\times$ precision lift over prevalence. This is competitive with the within-cohort TCGA models trained end-to-end on MSI labels (patient-level AUC 0.77–0.84 [37]; our bootstrap 95% CI reaches 0.78), and below the ~ 0.95 of clinical-grade detectors trained on large multi-institutional cohorts with full MSI supervision and external validation [39]. Astra reaches this without any

of that task-specific machinery: it sees no MSI label or TCGA slide in training and fits only a named-gene linear probe, keeping an interpretability those end-to-end detectors lack. As argued for trastuzumab (Section 4), predicted expression is the shared substrate, so gains to the base model from aggregated spatial transcriptomics carry over to this task with no new MSI supervision.

Because the probe is linear and its genes are named, the prediction traces back to features that align with MSI-H biology. Hypermutated, neoantigen-rich MSI-H tumors recruit an inflamed immune infiltrate [40], and the most recurrently selected genes are exactly that program: myeloid and antigen-handling markers (CCL3, SPI1, C5AR1, CPVL, FGR), the tryptophan-catabolizing checkpoint gene IDO1, the cross-presenting dendritic-cell receptor XCR1, and MHC class II (HLA-DQB2) are each chosen in over 90% of outer folds (Figure 20). As a spatial sanity check on the same immune-versus-epithelial axis, Figure 21 contrasts the cytotoxic-effector gene PRF1 (perforin), dense across MSI-H tumors and largely absent from MSS, with the epithelial-identity gene CDX2, whose loss is highly specific for mismatch-repair deficiency [41] and which produces the mirror image.

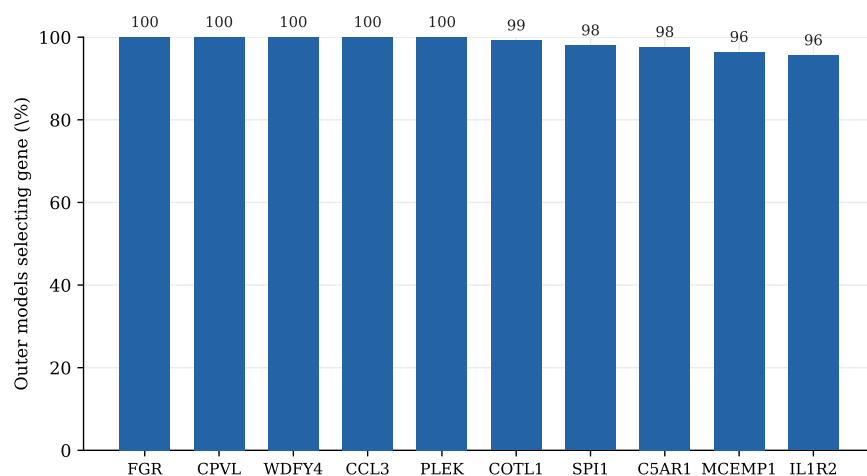


Figure 20: Top-10 genes by cross-validated selection frequency in the MSI-H probe (fraction of the 250 outer models retaining each gene).

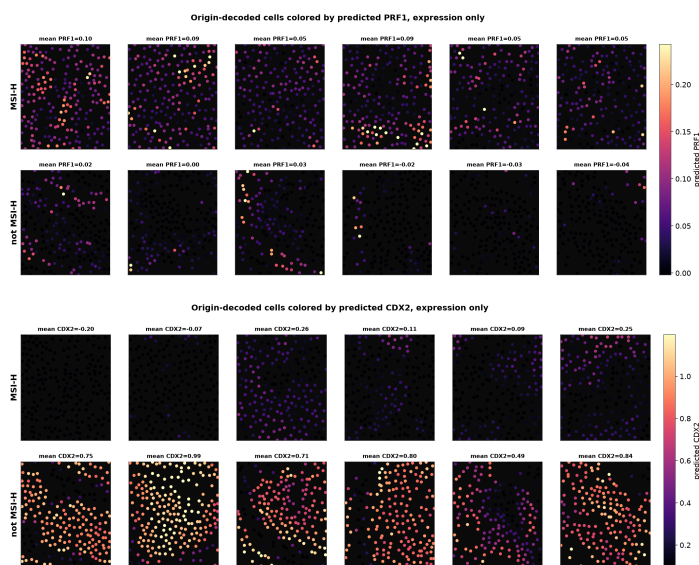


Figure 21: Astra-predicted PRF1 (top) and CDX2 (bottom) across MSI-H (top rows within each panel) and MSS (bottom rows) tumors.

A frozen, cross-domain expression model, without task-specific fine-tuning on MSI, therefore recovers an established clinical biomarker through an interpretable, named-gene route rather than an opaque classifier, and identifies the genes responsible. Two limits apply: these results are cross-validated within TCGA rather than externally validated, and the 33% MSI-H prevalence here is enriched relative to the roughly 15% observed clinically, so we present this as evidence rather than a deployable diagnostic.

7 Discussion

Astra establishes a general architecture for learning single-cell molecular representations from routine histology. By jointly inferring cellular position and expression, it removes the requirement for a spatial assay or separately matched segmentation at inference. The same frozen representation transfers from expression recovery to treatment-response prediction, survival stratification, and MSI-H pre-screening. Together, these findings suggest that spatial-expression supervision can produce histologic representations with clinical utility across treatment response, prognosis, and biomarker screening. Prospective, multi-institutional validation is required before deployment.

This separation between a general image-to-expression model and a lightweight task-specific model is important. A single frozen model could be run retrospectively over existing pathology archives, after which small labeled cohorts could be used to test response, prognosis, or biomarker hypotheses without retraining Astra for every endpoint. Single-cell output also preserves regional heterogeneity that slide-level embeddings and pseudobulk measurements can erase. In practice, this could support molecular-test triage, identify tissue regions for confirmatory profiling, and help prioritize patients for biomarker-directed studies while retaining a spatial record of the predicted signal.

One particularly relevant use is clinical-trial enrichment, which we raise as a hypothesis to be tested prospectively rather than a demonstrated capability. FDA guidance defines predictive enrichment as selecting patients “more likely to respond to the drug treatment than other patients with the condition” and notes that this can increase effect size and permit a smaller study population [42]. This matters because aggregate trial results can obscure spatially defined differences in treatment response. In NeoTRIP, adding atezolizumab did not significantly improve pathologic complete response overall (48.6% vs. 44.4%, $p = 0.48$), but retrospective imaging mass cytometry of archived trial tissue found that cellular composition and spatial organization distinguished immunotherapy-sensitive from resistant tumors [43]. In NSABP B-41, the H&E-derived DeSTIL spatial immune signature identified a subgroup with markedly better event-free survival on trastuzumab than the combination arm (HR 0.09; interaction $p = 0.024$), while signature-negative patients showed no treatment difference [44]. Exploratory analysis of KATHERINE likewise found no T-DM1 benefit in the subgroup with focal HER2 expression below 30% (HR 1.21; focal-versus-nonfocal interaction $p < 0.05$), although the subgroup was small and the confidence interval wide (0.58–2.51) [45]. These retrospective results show how an aggregate trial effect can conceal treatment-relevant spatial organization. An inexpensive H&E-based pre-screen for a response-associated spatial program could prioritize patients for confirmatory testing or enrollment, potentially recovering efficacy signals that an unselected trial would dilute. Its near-term value is as a scalable screening layer that reduces the samples needing costly molecular assays, not as a determinant of eligibility on its own. More broadly, Astra shows how inferred single-cell expression can turn routine histology into a scalable substrate for biomarker discovery, trial enrichment, and clinically directed molecular profiling.

8 Methods

8.1 Model architecture

Astra maps an H&E patch and a set of cell positions to per-cell gene expression. Each patch is a 256×256 -pixel tile resampled to $0.5 \mu\text{m}/\text{pixel}$ (a $128 \mu\text{m}$ field of view), then bilinearly resized

to 224×224 and ImageNet-normalized before the encoder. The image encoder is a frozen ProViGigaPath ViT-G/16 [46] with embedding dimension 1536; we keep its full token grid, 197 tokens per patch (one class token and a 14×14 grid of patch tokens), rather than pooling to the class token.

The shared trunk is a four-layer pre-norm transformer over these 197 tokens with eight attention heads (head dimension 192). Each layer normalizes with root-mean-square norm (RMSNorm [47]), $\text{RMSNorm}(x) = x / \sqrt{\frac{1}{d} \sum_i x_i^2 + \epsilon} \odot \gamma$, and uses a gated SwiGLU feed-forward block [48], $\text{SwiGLU}(x) = W_{\text{down}}(\text{SiLU}(W_{\text{gate}}x) \odot W_{\text{up}}x)$ with inner width 4,096, in place of a GELU MLP. Token position is injected by two-dimensional rotary position embeddings (RoPE [49]) applied to the queries and keys: each head’s channels are split in half, one half rotated by the token’s grid column and the other by its row (base frequency $\theta = 100$; the class token is given the identity rotation). Before rotation, queries and keys are per-head RMSNorm-normalized (QK-norm) for training stability. Concretely, for query/key vector u at grid position with rotary tables $(\cos, \sin)$, $\text{RoPE}(u) = u \odot \cos + \text{rot}_{\text{half}}(u) \odot \sin$, and attention is the usual softmax over $q^\top k / \sqrt{192}$. A final RMSNorm is applied before the heads read from the trunk.

Cells enter as queries. A cell at pixel (x, y) inside the patch is expressed as a patch-relative coordinate $(dx, dy) \in [-1, 1]^2$ and embedded with a NeRF-style Fourier feature map [50] using eight frequency bands $\omega_k = \pi 2^k$ for $k = 0, \dots, 7$:

$$\phi(dx, dy) = [\sin(\omega_k dx), \cos(\omega_k dx), \sin(\omega_k dy), \cos(\omega_k dy)]_{k=0}^7 \in \mathbb{R}^{32},$$

which a linear layer projects to 1536 and an RMSNorm normalizes. This query cross-attends (eight heads) to that patch’s 197 image tokens, producing one 1536-dimensional feature per cell; the feature is RMSNorm’d and refined by a residual SwiGLU feed-forward block, $\text{SwiGLU}(x) = W_{\text{down}}(\text{SiLU}(W_{\text{gate}}x) \odot W_{\text{up}}x)$ with inner width 4,096. Because each cell is queried at its own centroid, there is no nucleus-to-transcript matching step.

Three linear heads read the per-cell feature. The expression head is a linear map to the 1,656-gene panel with no output activation. The gene-presence (detector) head is a parallel linear map to 1,656 logits predicting whether each gene is non-zero in that cell. The density head is separate: it reshapes the 14×14 patch tokens to a spatial grid, applies a 1×1 projection to 192 channels, upsamples three times ($2 \times$ bilinear, 3×3 convolution, GELU) to 112×112 , and applies a softplus to emit a non-negative density map.

8.2 Targets, loss, and optimization

Expression targets are library-size normalized to a fixed total count of 200 per cell and then $\log(1 + x)$ -transformed; all recovery metrics and downstream features are computed in this log-normalized space. Supervision is restricted to the genes each Xenium slide actually measures through a per-cell presence mask, so a slide never penalizes genes outside its panel. The training objective sums three terms,

$$\mathcal{L} = \lambda_{\text{expr}} \text{MSE}(\hat{y}, y) + \lambda_{\text{pres}} \text{BCE}_w(\hat{p}, \mathbf{1}[y > 0]) + \lambda_{\text{dens}} \text{MSE}_\Sigma(\hat{D}, D) + \lambda_{\text{cnt}} |\sum \hat{D} - N|,$$

with $\lambda_{\text{expr}} = 100$, $\lambda_{\text{pres}} = 1$ (positive class weight 4), $\lambda_{\text{dens}} = 1000$, and $\lambda_{\text{cnt}} = 0.1$. The expression term is mean-squared error in log-normalized space (the optional Pearson term carried by the `mse_pcc` loss is disabled). The density target D places a unit-mass Gaussian ($\sigma = 2$ density pixels) at each true centroid, scored by summed MSE, and a companion L_1 term matches the integrated

density to the true cell count N . The model trains for 12 epochs with AdamW [51] (learning rate 7×10^{-4} , $\beta = (0.9, 0.95)$, weight decay 10^{-4} , gradient clip 1.0, batch size 48) under a 5% linear warmup followed by cosine decay to 10^{-6} . Patches are augmented with random flips, 90° rotations, light HED and color jitter; the encoder stays frozen throughout.

8.3 Gene panel and training data

Astra is trained on HEST-1k Xenium slides [52] across the target tissues. The per-tissue Xenium panels do not agree, so all measured genes are indexed into one universal axis and the 1,656-gene output is the union of genes measured anywhere in the corpus; each cell is supervised only on the genes its own slide measured. Patient-level cross-validation withholds entire slides per fold, and the 12 held-out slides used for the recovery analyses are disjoint from the 47 training slides.

8.4 Whole-slide preprocessing

For every downstream cohort, slides are tiled on a non-overlapping grid of 256-pixel tiles at $0.5 \mu\text{m}/\text{pixel}$, matching the training geometry. Each tile is stain-normalized by luminosity standardization: the tile is converted RGB to LAB, the 95th percentile p of the L (luminance) channel is taken, L is rescaled as $L' = \text{clip}(255 L/p, 0, 255)$ with the A and B channels unchanged, and the tile is converted back to RGB. This equalizes brightness per tile without a reference slide and leaves hue and saturation untouched.

Tiles are selected per task. For the TCGA-BRCA survival and MSI cohorts we rank tiles by tissue content and keep the top 1,000 per slide, where tissue is a mask of saturated, non-bright pixels (HSV saturation > 25 and value < 245) cleaned by 5×5 morphological opening and closing, and a tile’s score is the fraction of its pixels inside that mask. For the HER2 cohort we instead rasterize the pathologist tumor-ROI polygons and keep tiles whose ROI overlap is at least 0.5 and whose tissue fraction is at least 0.1, ranked by ROI coverage and capped at 512 tiles per slide.

8.5 Inference: cell decoding and expression

At inference Astra receives only H&E. For each tile the density head emits a non-negative map $\hat{D} \in \mathbb{R}_{\geq 0}^{H_d \times W_d}$ with $H_d = W_d = 112$ spanning the tile’s $128 \mu\text{m}$ field of view, so one density pixel is 2.29 source pixels ($1.14 \mu\text{m}$). Because the training target places a unit-mass Gaussian at every centroid, the integral of the map estimates the cell count,

$$\hat{N} = \sum_u \hat{D}_u, \quad C = \text{round}(\hat{N}),$$

and a pixel index $u = (r, c)$ maps to a patch-relative query coordinate by

$$\pi(u) = \left(\frac{2(c+\frac{1}{2})}{W_d} - 1, \frac{2(r+\frac{1}{2})}{H_d} - 1 \right) \in [-1, 1]^2.$$

Converting $\hat{D}$ into discrete centroids uses two decoders with complementary failure modes, selected per tile.

Peak decoding (non-maximum suppression). Let $\mathcal{N}_w(u)$ be the $w \times w$ box centered at u (zero-padded outside the map) and τ a peak-height floor. A pixel is a peak when it attains the maximum of its own neighborhood and clears the floor,

$$\mathcal{P} = \left\{ u : \hat{D}_u = \max_{v \in \mathcal{N}_w(u)} \hat{D}_v \text{ and } \hat{D}_u > \tau \right\},$$

and the decoded set is $\{\pi(u) : u \in \mathcal{P}\}$, optionally truncated to the C tallest peaks. We use $\tau = 0.005$ and $w = 3$. Two peaks at Chebyshev distance 1 would each have to dominate the other, so distinct peaks are at least 2 density pixels ($\approx 2.3 \mu\text{m}$) apart, which sets the method’s resolution limit. Its cardinality $|\mathcal{P}|$ is emergent rather than prescribed: where neighboring Gaussians merge into a single plateau the peak count falls below C , so peak decoding undercounts in the most crowded regions.

Count-matched centroid decoding (Lloyd / centroidal Voronoi). The second decoder fixes the cardinality at C and solves only for position. Let $S = \{u : \hat{D}_u > \tau\}$ be the support, $p_u = \pi(u)$, and normalize the densities into a distribution on S ,

$$w_u = \hat{D}_u / \sum_{v \in S} \hat{D}_v, \quad \sum_{u \in S} w_u = 1.$$

We seek C sites $Z = \{z_1, \dots, z_C\}$ minimizing the density-weighted quantization energy

$$F(Z) = \sum_{u \in S} w_u \min_{1 \leq k \leq C} \|p_u - z_k\|_2^2,$$

which Lloyd’s algorithm attacks by alternating a Voronoi assignment with a weighted-centroid update. Writing $a(u) = \arg \min_k \|p_u - z_k\|_2$ and $R_k = \{u \in S : a(u) = k\}$, the update is

$$z_k \leftarrow \frac{\sum_{u \in R_k} w_u p_u}{\sum_{u \in R_k} w_u} \quad (R_k \neq \emptyset; \text{ empty regions hold } z_k \text{ fixed}).$$

Each half-step is non-increasing in F : the assignment minimizes F over a at fixed Z , and the weighted mean is the minimizer of weighted squared distance at fixed a . So F descends monotonically to a local minimum whose stationary point is the centroidal-Voronoi condition, every site sitting at the density-weighted centroid of the region it owns. Sites are initialized at support pixels drawn from $\text{Categorical}(w)$ (without replacement when $|S| \geq C$), which starts them inside the mass and keeps the iteration short. We stop at a tolerance of 10^{-5} on $\max_k \|\Delta z_k\|_\infty$ or a cap of 8 iterations, and return exactly C centroids. Unlike peak decoding this cannot undercount, but its points are centroids of mass rather than nuclei, so it is less accurate wherever peaks are genuinely resolvable.

Adaptive selection. The decoders are combined per tile by testing how much of the predicted count the peaks can account for. With $P = |\mathcal{P}|$ the uncapped peak count and a saturation fraction $s = 0.8$, peaks are trusted when $P \geq sC$ and the centroid decoder is used otherwise; $P \ll C$ is the signature of merged blobs. The ratio P/C is thus an estimate of the fraction of cells the map resolves as distinct maxima.

Algorithm 1 Adaptive density decoding for one tile

Require: map $\hat{D}$, floor $\tau = 0.005$, window $w = 3$, saturation $s = 0.8$, iteration cap 8

- 1: $C \leftarrow \text{round}(\sum_u \hat{D}_u)$ ▷ count from the integrated density
- 2: $\mathcal{P} \leftarrow \{u : \hat{D}_u = \max_{v \in \mathcal{N}_w(u)} \hat{D}_v \text{ and } \hat{D}_u > \tau\}$
- 3: **if** $C > 0$ **and** $|\mathcal{P}| \geq sC$ **then**
- 4: **return** $\pi(u)$ for the $\min(C, |\mathcal{P}|)$ tallest $u \in \mathcal{P}$ ▷ peaks resolve the count
- 5: $S \leftarrow \{u : \hat{D}_u > \tau\}$; $w_u \leftarrow \hat{D}_u / \sum_{v \in S} \hat{D}_v$
- 6: $z_k \leftarrow \pi(u_k)$ with $u_k \sim \text{Categorical}(w)$, $k = 1, \dots, C$ ▷ peaks saturated
- 7: **for** $t = 1, \dots, 8$ **do**
- 8: $a(u) \leftarrow \arg \min_k \|\pi(u) - z_k\|_2$ for all $u \in S$
- 9: $z'_k \leftarrow (\sum_{u: a(u)=k} w_u \pi(u)) / (\sum_{u: a(u)=k} w_u)$ for non-empty regions
- 10: **if** $\max_k \|z'_k - z_k\|_\infty < 10^{-5}$ **then** $z \leftarrow z'$; **break**
- 11: $z \leftarrow z'$
- 12: **return** $z_1, \dots, z_C$

The decoded centroids are used directly as the cell queries of Section 8.1: each is Fourier-embedded and cross-attends to its own tile’s 197 image tokens, giving an $[n_{\text{cells}} \times 1656]$ matrix of predicted log-normalized expression per tile, and each is offset from its tile anchor into absolute slide coordinates so cells pool across tiles. The gene-presence probability, when used, is thresholded at a single value chosen on held-out validation slides to zero out predicted absences.

8.6 Spatial-recovery evaluation

Cell matching. The decoded cells and the measured Xenium cells are independent point sets of sizes M and N , so scoring first pairs them by greedy nearest-neighbor matching. Let $d_{ij} = \|c_i^{\text{true}} - c_j^{\text{pred}}\|_2$ be pairwise distances and τ the gate, set to the median nearest-neighbor spacing among measured cells. Candidate pairs with $d_{ij} \leq \tau$ are sorted ascending and consumed greedily, each cell used once:

Algorithm 2 Greedy one-to-one cell matching

- 1: $\mathcal{C} \leftarrow \{(i, j) : d_{ij} \leq \tau\}$ sorted by increasing d_{ij}
- 2: $\mathcal{M} \leftarrow \emptyset$; $\text{used}_{\text{true}} \leftarrow \emptyset$; $\text{used}_{\text{pred}} \leftarrow \emptyset$
- 3: **for** $(i, j) \in \mathcal{C}$ **do**
- 4: **if** $i \notin \text{used}_{\text{true}}$ **and** $j \notin \text{used}_{\text{pred}}$ **then**
- 5: $\mathcal{M} \leftarrow \mathcal{M} \cup \{(i, j)\}$; mark i, j used
- 6: **return** matched pairs $\mathcal{M}$ ▷ unmatched cells are misses / false positives

All per-gene correlations below are computed on the matched expression matrices $Y^{\text{true}}, \hat{Y} \in \mathbb{R}^{|\mathcal{M}| \times G}$ obtained by reindexing to $\mathcal{M}$.

Per-gene Spearman. Spearman’s ρ is Pearson’s r applied to ranks. For gene g , let $r(\cdot)$ assign average ranks (ties averaged) down the column of matched cells and $\bar{r}$ its mean:

$$a = r(\hat{Y}_{:,g}), \quad b = r(Y_{:,g}^{\text{true}}), \quad \rho_g = \frac{\sum_c (a_c - \bar{a})(b_c - \bar{b})}{\sqrt{\sum_c (a_c - \bar{a})^2} \sqrt{\sum_c (b_c - \bar{b})^2}}.$$

A gene with zero variance in either column is undefined and dropped from the aggregate; the headline number is the mean (or median) of ρ_g over kept genes. The out-of-distribution 10x slides instead report per-gene Pearson r on the matched values directly (no ranking).

Moran’s I. Spatial structure per gene uses a k -nearest-neighbor graph over cell coordinates ($k = 6$), row-normalized to weights W with $\sum_j W_{ij} = 1$ (so $W_{ij} = 1/k$ for the k neighbors of i , else 0). For gene g with expression x , center it as $z = x - \bar{x}$. Since row-normalization makes $S_0 = \sum_{ij} W_{ij} = N$, the leading N/S_0 factor is 1 and Moran’s I is the normalized quadratic form

$$I_g = \frac{z^\top W z}{z^\top z} = \frac{\sum_i z_i (W z)_i}{\sum_i z_i^2}, \quad (W z)_i = \frac{1}{k} \sum_{j \in \text{knn}(i)} z_j.$$

$(W z)_i$ is cell i ’s neighbor-averaged deviation, so the numerator rewards cells whose own deviation agrees in sign with their neighborhood: $I_g \rightarrow 1$ is smooth spatial clustering, $I_g \approx 0$ is spatially random.

Aggregation by spatial strength. Genes are ranked by I_g computed on *measured* expression, so selection never sees the prediction. Writing π for that descending order, the top-50 boxplot per slide is $\{\rho_{\pi(1)}, \dots, \rho_{\pi(50)}\}$ (box = quartiles Q_1, Q_3 , line = median, Tukey whiskers to the farthest point within $[Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}]$), and the recovery-versus-gene-count curve is the running cumulative mean

$$S(n) = \frac{1}{n} \sum_{t=1}^n \rho_{\pi(t)},$$

averaged across slides within each tissue.

8.7 Downstream clinical models

The frozen Astra model is never fine-tuned on any clinical label; only a lightweight probe is trained per task. All fitting happens strictly inside each cross-validation training split.

Patient representations. For survival and MSI we pseudobulk: with per-tile decoded expression $\hat{Y}^{(t)} \in \mathbb{R}^{n_t \times G}$ over retained tiles $t = 1, \dots, T$, the patient vector averages over *all* cells,

$$u = \frac{\sum_t \sum_{c=1}^{n_t} \hat{Y}_{c,:}^{(t)}}{\sum_t n_t} \in \mathbb{R}^G, \quad G = 1,656.$$

For HER2 we keep heterogeneity instead. Each tile is collapsed to its mean profile $m^{(t)} = \frac{1}{n_t} \sum_c \hat{Y}_{c,:}^{(t)}$, transformed as $\tilde{m}^{(t)} = \log(1 + \max(m^{(t)}, 0))$, and each gene g is summarized across the T tiles by two dispersion features,

$$\text{STD}_g = \sqrt{\frac{1}{T} \sum_t (\tilde{m}_g^{(t)} - \bar{\tilde{m}}_g)^2}, \quad \text{GAP}_g = Q_{90}(\tilde{m}_g^{(\cdot)}) - Q_{50}(\tilde{m}_g^{(\cdot)}),$$

giving a $2G = 3,312$ -dimensional feature vector per patient that excludes mean abundance.

Survival (TCGA-BRCA, luminal). With right-censored overall survival (t_i, δ_i) we fit a Cox proportional-hazards model $h_i(t) = h_0(t) \exp(\beta^\top x_i)$ by maximizing the partial likelihood over the risk sets $R(t_i) = \{j : t_j \geq t_i\}$,

$$\ell(\beta) = \sum_{i:\delta_i=1} \left[\beta^\top x_i - \log \sum_{j \in R(t_i)} e^{\beta^\top x_j} \right].$$

Within each training fold we standardize genes, rank them by univariate association (each gene’s single-covariate Cox score), and keep the top K , with K chosen per fold by an inner 3-fold cross-validation over the development patients (median 40); the multivariable model is then fit on those genes. The risk score for a held-out patient is $\hat{\beta}^\top x$; concordance is Harrell’s C , the fraction of comparable patient pairs whose predicted risk orders agree with survival:

$$C = \frac{\sum_{i:\delta_i=1} \sum_{j:t_j > t_i} \mathbf{1}[\hat{\beta}^\top x_i > \hat{\beta}^\top x_j]}{\sum_{i:\delta_i=1} \sum_{j:t_j > t_i} 1}.$$

This is repeated over 5-fold cross-validation (50 repeats); out-of-fold scores are pooled, and patients are compared by a two-sided log-rank test both at the median split and between the top and bottom risk quartiles.

Response and MSI (logistic probe). Both binary tasks fit an L2-regularized logistic regression on standardized features,

$$\min_{w,b} \frac{1}{2} \|w\|_2^2 + C \sum_i \log(1 + e^{-y_i (w^\top x_i + b)}), \quad \hat{p}_i = \sigma(w^\top x_i + b),$$

with $y_i \in \{-1, +1\}$. MSI (TCGA-COAD/READ) fits this on the pseudobulk vector against the genomic MANTIS label under nested cross-validation: a repeated stratified 5-fold outer loop (50 repeats), with the number of AUC-ranked genes chosen by an inner 5-fold loop inside each training split (median 160 genes) before refitting on the full development set and scoring the held-out fold.

HER2 response adds nested feature-count selection so the dispersion features and their number are chosen without touching test patients:

Algorithm 3 HER2 response: nested inner-CV feature-count selection (one outer fold)

- 1: split cohort $\rightarrow$ 17 test, 68 development (stratified by response)
 - 2: **for** $K = 1, \dots, 20$ **do**
 - 3: **for** each inner stratified 5-fold split of the 68 development patients **do**
 - 4: standardize on the inner-train fold; rank genes by $|\text{corr}(\text{feature}, y)|$
 - 5: fit logistic regression on the top- K features; predict the inner-validation fold
 - 6: record the pooled inner-fold AUROC for K
 - 7: $K^* \leftarrow \arg \max_K$ pooled inner-fold AUROC
 - 8: refit top- K^* model on all 68 development patients
 - 9: predict $\hat{p}$ for the 17 test patients
-

The full procedure runs over 50 repeats of stratified 5-fold splits; the 250 outer models yield out-of-fold probabilities that are averaged per patient before scoring, and 95% confidence intervals come from 2,000 bootstrap resamples of the patients. Feature-selection frequencies (e.g. how often each dispersion feature enters the top- K^*) are tallied across the 250 models.

References

- [1] The emerging landscape of spatial profiling technologies. *Nature Biotechnology*, 2024. <https://www.nature.com/articles/s41587-024-02543-2>.
- [2] Spatial transcriptomics in pathology. *The American Journal of Pathology*, 2024. [https://ajp.amjpathol.org/article/S0002-9440\(24\)00276-1/fulltext](https://ajp.amjpathol.org/article/S0002-9440(24)00276-1/fulltext).
- [3] Yusuke Ogitani et al. Bystander killing effect of ds-8201a, a novel anti-her2 antibody–drug conjugate, in tumors with heterogeneous her2 expression. *Cancer Science*, 107:1039–1046, 2016. <https://doi.org/10.1111/cas.12966>.
- [4] Anna H. Staudacher and Michael P. Brown. Antibody drug conjugates and bystander killing: is antigen-dependent internalisation required? *British Journal of Cancer*, 117:1736–1742, 2017. <https://doi.org/10.1038/bjc.2017.367>.
- [5] Shanu Modi et al. Trastuzumab deruxtecan in previously treated her2-low advanced breast cancer (destiny-breast04). *New England Journal of Medicine*, 387:9–20, 2022. <https://doi.org/10.1056/NEJMoa2203690>.
- [6] Daniel S. Chen and Ira Mellman. Elements of cancer immunity and the cancer–immune set point. *Nature*, 541:321–330, 2017. <https://doi.org/10.1038/nature21349>.
- [7] Rita Cabrita et al. Tertiary lymphoid structures improve immunotherapy and survival in melanoma. *Nature*, 577:561–565, 2020. <https://doi.org/10.1038/s41586-019-1914-8>.
- [8] Beth A. Helmink et al. B cells and tertiary lymphoid structures promote immunotherapy response. *Nature*, 577:549–555, 2020. <https://doi.org/10.1038/s41586-019-1922-8>.
- [9] Bryan He et al. Integrating spatial gene expression and breast tumour morphology via deep learning (st-net). *Nature Biomedical Engineering*, 2020. <https://www.nature.com/articles/s41551-020-0578-x>.
- [10] Minxing Pang et al. Leveraging information in spatial transcriptomics to predict super-resolution gene expression from histology images (histogene). *bioRxiv*, 2021. <https://www.biorxiv.org/content/10.1101/2021.11.28.470212v1>.
- [11] Yuansong Zeng et al. Spatial transcriptomics prediction from histology jointly through transformer and graph neural networks (hist2st). *Briefings in Bioinformatics*, 2022. <https://academic.oup.com/bib/article/23/5/bbac297/6650011>.
- [12] Ghist: predicting single-cell spatial gene expression from histology. *Nature Methods*, 2025. <https://www.nature.com/articles/s41592-025-02795-z>.
- [13] Single-cell gene expression prediction from h&e images. *medRxiv*, 2025. <https://www.medrxiv.org/content/10.1101/2025.02.09.25321567v2>.
- [14] Motoki Takaku, Sara A. Grimm, and Paul A. Wade. Gata3 in breast cancer: Tumor suppressor or oncogene? *Gene Expression*, 2015. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4758516/>.
- [15] Wataru Nishie. Collagen xvii processing and blistering skin diseases. *Acta Dermato-Venereologica*, 2020. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9128997/>.

- [16] Yingjian You et al. Role of f-box factor foxj1 in differentiation of ciliated airway epithelial cells. *American Journal of Physiology-Lung Cellular and Molecular Physiology*, 2004. <https://doi.org/10.1152/ajplung.00170.2003>.
- [17] Hiroshi Kobayashi et al. The balance of stromal bmp signaling mediated by grem1 and islr drives colorectal carcinogenesis. *Gastroenterology*, 2021. <https://doi.org/10.1053/j.gastro.2020.11.011>.
- [18] Cd45 (ptprc): the leukocyte common antigen. *PMC*, 2021. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8380896/>.
- [19] Saman Farahmand et al. Deep learning trained on hematoxylin and eosin tumor region of interest predicts her2 status and trastuzumab treatment response in her2+ breast cancer. *Modern Pathology*, 2021. <https://doi.org/10.1038/s41379-021-00911-w>.
- [20] Daniel Rodrigues de Bastos, Adhemar Longatto-Filho, Mércia Patrícia Ferreira Conceição, and Lara Termini. High levels of superoxide dismutase 2 are associated with worse prognosis in patients with breast cancer. *European Journal of Breast Health*, 20:185–193, 2024. <https://doi.org/10.4274/ejbh.galenos.2024.2024-3-6>.
- [21] Maria José Oliveira, João G. Costa, and Ana S. Fernandes. Interaction between redox regulation, immune activation, and response to treatment in her2+ breast cancer. *Redox Biology*, 82:103609, 2025. <https://doi.org/10.1016/j.redox.2025.103609>.
- [22] Fernanda S. Ramos et al. Pdia3 and pdia6 gene expression as an aggressiveness marker in primary ductal breast cancer. *Genetics and Molecular Research*, 14:6960–6967, 2015. <https://doi.org/10.4238/2015.June.26.4>.
- [23] Anna Germon, Kate J. Heesom, Reiss Amoah, and Josephine C. Adams. Protein disulfide isomerase a3 activity promotes extracellular accumulation of proteins relevant to basal breast cancer outcomes in human mda-mb-a231 breast cancer cells. *American Journal of Physiology-Cell Physiology*, 324:C113–C132, 2023. <https://doi.org/10.1152/ajpcell.00445.2022>.
- [24] L. H. Zhang and X. Zhang. Grp78/bip and the unfolded protein response in her2 breast cancer and trastuzumab resistance. *Cancer Letters*, 330:123–129, 2013. <https://doi.org/10.1016/j.canlet.2012.11.014>.
- [25] T. Boulding et al. Cd44 and trastuzumab resistance in her2-positive breast cancer cells. *Breast Cancer Research and Treatment*, 152:489–500, 2015. <https://doi.org/10.1007/s10549-015-3414-3>.
- [26] Yuhua Zhao et al. Overcoming trastuzumab resistance in breast cancer by targeting the hsf1-ldha glycolysis axis. *Cancer Research*, 71:4585–4597, 2011. <https://doi.org/10.1158/0008-5472.CAN-11-0127>.
- [27] M. Balbin et al. Prognostic value of pepsinogen c (progastricsin) in breast cancer. *Journal of Clinical Oncology*, 13:54–61, 1995. <https://doi.org/10.1200/JCO.1995.13.1.54>.
- [28] Q. K. Y. Chan et al. Fstl1 promotes cancer stemness and chemoresistance through integrin/wnt signaling. *Cancer Biology & Therapy*, 19:1067–1077, 2018. <https://doi.org/10.1080/15384047.2018.1529101>.

- [29] Bradley M. Turner et al. Napsin a, a new marker for lung adenocarcinoma, and its diagnostic utility. *Archives of Pathology & Laboratory Medicine*, 136:163–171, 2012. <https://pubmed.ncbi.nlm.nih.gov/22288963/>.
- [30] Gene expression and benefit of chemotherapy in women with node-negative, estrogen receptor–positive breast cancer. *Journal of Clinical Oncology*, 2008. <https://doi.org/10.1200/JCO.2008.18.1370>.
- [31] Marcus Schmidt et al. Immunoglobulin kappa chain: a robust single-gene prognostic and predictive marker in breast cancer. *Clinical Cancer Research*, 18:2695–2703, 2012. <https://doi.org/10.1158/1078-0432.CCR-12-0566>.
- [32] Carsten Denkert et al. Tumour-infiltrating lymphocytes and prognosis in breast cancer. *Cancers*, 15:4479, 2023. <https://doi.org/10.3390/cancers15184479>.
- [33] Ning Wang et al. Secretory leukocyte protease inhibitor (slpi) as a target for inhibiting metastasis of triple-negative breast cancer. *Oncotarget*, 8:108292–108302, 2017. <https://doi.org/10.18632/oncotarget.22660>.
- [34] K. Zhang et al. Causal roles of cathepsin family members in breast cancer subtypes. *Discover Oncology*, 16:427, 2025. <https://doi.org/10.1007/s12672-025-03787-3>.
- [35] Voraratt Champattanachai et al. microRNA-200c/141 upregulates serpinb2 to promote breast cancer cell metastasis and reduce patient survival. *Oncotarget*, 8:24810–24827, 2017. <https://doi.org/10.18632/oncotarget.15680>.
- [36] Fda grants accelerated approval to pembrolizumab for the first tissue/site-agnostic indication, 2017. <https://www.fda.gov/drugs/resources-information-approved-drugs/fda-grants-accelerated-approval-pembrolizumab-first-tissue-site-agnostic-indication>.
- [37] Jakob Nikolas Kather et al. Deep learning can predict microsatellite instability directly from histology in gastrointestinal cancer. *Nature Medicine*, 2019. <https://www.nature.com/articles/s41591-019-0462-y>.
- [38] Esko A. Kautto et al. Performance evaluation for rapid detection of pan-cancer microsatellite instability with mantis. *Oncotarget*, 8:7452–7463, 2017. <https://doi.org/10.18632/oncotarget.13918>.
- [39] Amelie Echle et al. Clinical-grade detection of microsatellite instability in colorectal cancer by deep learning. *Gastroenterology*, 2020. <https://doi.org/10.1053/j.gastro.2020.06.021>.
- [40] The vigorous immune microenvironment of microsatellite instable colon cancer. *Cancer Discovery*, 2015. <https://aacrjournals.org/cancerdiscovery/article/5/1/43/4777/>.
- [41] Loss of cdx2 expression and microsatellite instability. *Frontiers in Oncology*, 2013. <https://www.frontiersin.org/journals/oncology/articles/10.3389/fonc.2013.00265/full>.
- [42] U.S. Food and Drug Administration. Enrichment strategies for clinical trials to support determination of effectiveness of human drugs and biological products: Guidance for industry, 2019. <https://www.fda.gov/media/121320/download>.
- [43] Xiao Qian Wang et al. Spatial predictors of immunotherapy response in triple-negative breast cancer. *Nature*, 621:868–876, 2023. <https://doi.org/10.1038/s41586-023-06498-3>.

- [44] Satvika Bharadwaj et al. Computationally derived spatial immune signature identifies trastuzumab responders in her2-positive breast cancer: Nsabp b-41 clinical trial validation. *Clinical Cancer Research*, 2026. <https://doi.org/10.1158/1078-0432.CCR-25-4185>.
- [45] Carsten Denkert et al. Biomarker data from the phase iii katherine study of adjuvant t-dm1 versus trastuzumab for residual invasive disease after neoadjuvant therapy for her2-positive breast cancer. *Clinical Cancer Research*, 29:1569–1581, 2023. <https://doi.org/10.1158/1078-0432.CCR-22-1989>.
- [46] Hanwen Xu et al. A whole-slide foundation model for digital pathology from real-world data (prov-gigapath). *Nature*, 630:181–188, 2024. <https://www.nature.com/articles/s41586-024-07441-w>.
- [47] Biao Zhang and Rico Sennrich. Root mean square layer normalization. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. <https://arxiv.org/abs/1910.07467>.
- [48] Noam Shazeer. Glu variants improve transformer, 2020. <https://arxiv.org/abs/2002.05202>.
- [49] Jianlin Su et al. Roformer: enhanced transformer with rotary position embedding. *Neurocomputing*, 568, 2024. <https://arxiv.org/abs/2104.09864>.
- [50] Matthew Tancik et al. Fourier features let networks learn high frequency functions in low dimensional domains. *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. <https://arxiv.org/abs/2006.10739>.
- [51] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019. <https://arxiv.org/abs/1711.05101>.
- [52] Guillaume Jaume et al. Hest-1k: a dataset for spatial transcriptomics and histology image analysis. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. <https://arxiv.org/abs/2406.16192>.